

Datatype Mapping: CLR / SQL Server / Oracle / PostgreSQL

Schema-conversion reference — defaults mirror the TflOmniDb provider type mappers.

CLR / .NET	SQL Server	Oracle	PostgreSQL	Notes
Boolean & integer				
bool	BIT	NUMBER(1,0)	BOOLEAN	Oracle <23c: NUMBER(1) 0/1; 23c+ has BOOLEAN
byte	TINYINT	NUMBER(3,0)	SMALLINT	TINYINT is unsigned 0-255; PG has no 1-byte int
short / Int16	SMALLINT	NUMBER(5,0)	SMALLINT	
int / Int32	INT	NUMBER(10,0)	INTEGER	
long / Int64	BIGINT	NUMBER(19,0)	BIGINT	
long (identity)	INT IDENTITY	GENERATED AS IDENTITY	GENERATED AS IDENTITY / serial	Oracle <12c: SEQUENCE + trigger
Exact numeric				
decimal	DECIMAL(p,s)	NUMBER(p,s)	NUMERIC(p,s)	NUMERIC == DECIMAL
decimal	NUMERIC(p,s)	NUMBER(p,s)	NUMERIC(p,s)	Oracle NUMBER max precision 38
decimal	MONEY	NUMBER(19,4)	NUMERIC(19,4)	Avoid PG MONEY (locale); canonical NUMERIC(19,4)
decimal	SMALLMONEY	NUMBER(10,4)	NUMERIC(10,4)	Range ~ +/-214,748.3647
Approximate numeric				
float / Single	REAL	BINARY_FLOAT	REAL	REAL = FLOAT(24), 4-byte
double / Double	FLOAT	BINARY_DOUBLE	DOUBLE PRECISION	FLOAT(53) = 8-byte double; FLOAT(24)->REAL
Character & text				
string	VARCHAR(n)	VARCHAR2(n CHAR)	VARCHAR(n)	Oracle VARCHAR2 <= 4000 bytes (32767 extended)
string	NVARCHAR(n)	NVARCHAR2(n)	VARCHAR(n)	Unicode; PG is UTF-8 native (no N-prefix)
string	CHAR(n)	CHAR(n CHAR)	CHAR(n)	Fixed length, space-padded
string	NCHAR(n)	NCHAR(n)	CHAR(n)	Unicode fixed length
string	VARCHAR(MAX)	CLOB	TEXT	Large ANSI text (> 8000 bytes)
string	NVARCHAR(MAX)	NCLOB	TEXT	Large Unicode text
string	TEXT (legacy)	CLOB	TEXT	Deprecated -> use VARCHAR(MAX)
string	NTEXT (legacy)	NCLOB	TEXT	Deprecated -> use NVARCHAR(MAX)
Date & time				
DateTime / DateOnly	DATE	DATE	DATE	Oracle DATE INCLUDES time-of-day (to sec)
TimeSpan / TimeOnly	TIME(p)	INTERVAL DAY TO SECOND	TIME(p)	Oracle has no TIME-of-day type
DateTime	DATETIME2(p)	TIMESTAMP(p)	TIMESTAMP(p)	Preferred date+time; p = frac-sec precision
DateTime	DATETIME (legacy)	TIMESTAMP(3)	TIMESTAMP(3)	~3.33 ms precision; range 1753-9999

CLR / .NET	SQL Server	Oracle	PostgreSQL	Notes
DateTime	SMALLDATETIME	TIMESTAMP(0)	TIMESTAMP(0)	Minute precision; 1900-2079
DateTimeOffset	DATETIMEOFFSET	TIMESTAMP WITH TIME ZONE	TIMESTAMPTZ	Timezone-aware
Binary				
byte[]	VARBINARY(n)	RAW(n)	BYTEA	Oracle RAW <= 2000 bytes; beyond -> BLOB
byte[]	BINARY(n)	RAW(n)	BYTEA	Fixed; PG has no fixed-length binary
byte[]	VARBINARY(MAX)	BLOB	BYTEA	Large binary object
byte[]	IMAGE (legacy)	BLOB	BYTEA	Deprecated -> use VARBINARY(MAX)
Identifier & concurrency				
Guid	UNIQUEIDENTIFIER	RAW(16)	UUID	Oracle SYS_GUID(); PG native UUID
byte[] (rowversion)	ROWVERSION	(none)	(xmin sys col)	No portable equiv -> app-managed int version
XML & JSON				
string / SqlXml	XML	XMLTYPE	XML	Native XML on all three
string	JSON / NVARCHAR(MAX)	JSON (21c+)/CLOB	JSONB	PG JSONB preferred; Oracle <21c CLOB + IS JSON
No direct equivalent (redesign)				
object	SQL_VARIANT	(none / ANYDATA)	(none)	Redesign to a typed column
(tree)	HIERARCHYID	(none)	LTREE (extension)	Remodel as a materialized path
(spatial)	GEOGRAPHY / GEOMETRY	SDO_GEOMETRY	geometry/geography	Oracle Spatial / PostGIS extension

General notes

- Defaults mirror the TfiOmniDb provider type mappers; per-column overrides exist via the [DbColumn] SqlServerType / OracleType / PostgresType escape hatches.
- Money is deliberately NUMERIC(19,4) on all three engines - PostgreSQL's MONEY type is locale-dependent and unsafe for financial code.
- Identifier casing: Oracle folds unquoted identifiers to UPPER, PostgreSQL to lower. Quote consistently, or pick one case across the schema.
- Oracle DATE always carries a time component (to seconds). Use TIMESTAMP for fractional seconds; for pure dates, enforce truncation in the app.
- Booleans: target Oracle <23c as NUMBER(1) with 0/1 values; Oracle 23c+ and PostgreSQL have a native BOOLEAN.
- Auto-increment: PostgreSQL and Oracle 12c+ support GENERATED ... AS IDENTITY; older Oracle needs a SEQUENCE + trigger.
- Row versioning: there is no portable equivalent to SQL Server ROWVERSION. Use an app-managed integer version column (TfiOmniDb IsRowVersion) checked in the UPDATE WHERE clause; Oracle ORA_ROWSCN / PostgreSQL xmin are not drop-in replacements.
- LOBs map (N)VARCHAR(MAX)/VARBINARY(MAX) -> Oracle CLOB/NCLOB/BLOB -> PostgreSQL TEXT/BYTEA.
- PostgreSQL is UTF-8 native, so NVARCHAR and VARCHAR both collapse to VARCHAR/TEXT; JSONB beats JSON for indexing (GIN); BYTEA covers all binary widths.